

This section contains five problems to be solved using integration by parts. Show all work for each problem in order to receive full credit. Clearly label your selections for u and dv and circle your final answer.

- 1) $\int x e^{4x} dx$
- 2) $\int x^2 e^{4x} dx$
- 3) $\int x \sin(3x) dx$
- 4) $\int (3x^2 - x) \sin(x) dx$
- 5) $\int e^{2x} \sin x dx$

Due to scheduling issues
 * Not done... I'm gonna keep
 working and submit later.
 for now... here are problems 1-4

This section contains five problems to be solved using the techniques covered in section 3.2 of the text and our notes. These “piles of trigonometric function problems” may require identities and techniques from precalculus to manipulate algebraically. Refer to page 626 and 627 of our .pdf text as needed.

- 6) $\int \sin^7(x) \cos(x) dx$
- 7) $\int \sin^7(x) \cos^2(x) dx$
- 8) $\int \tan(x) \sec^3(x) dx$
- 9) $\int \tan^3(x) \sec^4(x) dx$
- 10) $\int \tan^4(3x) \cos^6(3x) dx$

This final section contains five problems to be solved using the techniques covered in section 3.3 of the text and our notes. These will often turn into problems similar to those in the previous section so we get to double dip on 3.2 practice.

- 11) $\int \frac{1}{\sqrt{1+9x^2}} dx$
- 12) $\int \frac{x^3}{\sqrt{9-x^2}} dx$
- 13) $\int \frac{\sqrt{x^2-25}}{x} dx$

$$14) \int \frac{1}{\sqrt{4-x^2}} dx$$

$$15) \int \frac{1}{(1+x^2)^{\frac{3}{2}}} dx$$

16) Use integration by parts to create the reduction formula below for $n = 3$. You may wish to play around with your initial choices for u and dv and keep in mind that the correct solution to this process requires escaping an infinite loop of integration by parts creating a copy of the very integral we wish to solve. Please reach out if this one seems a bit weird, but I want to assure you this is absolutely what math is about.

$$\int \sin^n(x) dx = -\frac{1}{n} \sin^{n-1}(x) \cos(x) + \frac{n-1}{n} \int \sin^{n-2}(x) dx$$

$$1. \int x e^{4x} dx$$

$$u = x \quad dv = e^{4x} dx$$

$$du = dx \quad v = \int e^{4x} dx = \frac{1}{4} e^{4x}$$

$$\int u dv = uv - \int v du$$

$$\begin{aligned} \int x e^{4x} dx &= \frac{1}{4} x e^{4x} - \int \frac{1}{4} e^{4x} dx \\ &= \frac{1}{4} x e^{4x} - \frac{1}{4} \left(\frac{1}{4} e^{4x} \right) + C \\ &= \frac{1}{4} x e^{4x} - \frac{1}{16} e^{4x} + C \\ &= \frac{1}{16} e^4 (4x - 1) + C \end{aligned}$$

$$2. \int x^2 e^{4x} dx$$

$$u = x^2 \quad dv = e^{4x} dx$$

$$du = 2x dx \quad v = \int e^{4x} dx = \frac{1}{4} e^{4x}$$

$$\int u dv = uv - \int v du$$

$$\int u dv = uv - \int v du$$

$$\int x^2 e^{4x} dx = \frac{1}{4} x^2 e^{4x} - \int \frac{1}{4} e^{4x} (2x dx)$$

$$\frac{1}{4} x^2 e^{4x} - \frac{1}{2} x e^{4x} dx$$

/

$$u_2 = x \quad \checkmark \quad dv_2 = e^{4x} dx$$

$$du_2 = dx \quad v_2 = \frac{1}{4} e^{4x}$$

$$\int x e^{4x} dx = \frac{1}{4} x e^{4x} - \int \frac{1}{4} e^{4x} dx$$

$$= \frac{1}{4} x e^{4x} - \frac{1}{16} e^{4x}$$

$$\int x^2 e^{4x} dx = \frac{1}{4} x^2 e^{4x} - \frac{1}{2} \left(\frac{1}{4} x e^{4x} - \frac{1}{16} e^{4x} \right) + C$$

$$\frac{1}{4} x^2 e^{4x} - \frac{1}{8} x e^{4x} - \frac{1}{32} e^{4x} + C$$

$$\frac{1}{32} e^{4x} (8x^2 - 4x + 1) + C$$

$$3. \int x \sin(3x) dx$$

$$u = x \quad dv = \sin(3x) dx$$

$$du = dx \quad v = -\frac{1}{3} \cos(3x)$$

Chain rule
 $\frac{d}{dx} [f(g(x))] = f'(g(x)) g'(x)$

$$\int u dv = uv - \int v du$$

$$-\frac{1}{3} x \cos(3x) - \int -\frac{1}{3} \cos(3x) dx$$

$$\int x \sin(3x) dx = -\frac{1}{3} x \cos 3x + \frac{1}{9} \int \cos 3x dx$$

$$\int \cos 3x dx = \frac{1}{3} \sin(3x)$$

$$-\frac{1}{3}x \cos 3x + \frac{1}{3}\left(\frac{1}{3} \sin 3x\right)$$

$$\int x \sin(3x) dx = -\frac{1}{3}x \cos 3x + \frac{1}{9} \sin 3x + C$$

$$4. \int (3x^2 - x) \sin(x) dx$$

$$u = 3x^2 - x \quad dv = \sin x dx$$

$$du = (6x - 1) dx \quad v = -\cos x$$

$$\int u dv = uv - \int v du$$

$$\int (3x^2 - x) \sin x dx = (3x^2 - x)(-\cos x) - \int (-\cos x)(6x - 1) dx$$

$$- (3x^2 - x)(\cos x) + \int (6x - 1) \cos x dx$$

$$\int (6x - 1) \cos x dx$$

$$u_2 = 6x - 1 \quad dv_2 = \cos x dx$$

$$du_2 = 6 dx \quad v_2 = \sin x$$

$$\int (6x - 1) \cos x dx = (6x - 1) \sin x - \int 6 \sin(x) dx$$

$$\int 6 \sin x dx = -6 \cos x$$

$$\int (6x - 1) \cos x dx = (6x - 1) \sin x - (-6 \cos(x))$$

$$(6x - 1) \sin x + 6 \cos x$$

$$\int (3x^2 - x) \sin x \, dx = -(3x^2 - x) \cos x + (6x - 1) \sin x + 6 \cos x + C$$

$$(-3x^2 + x + 6 \cos x) + (6x - 1) \sin x + C$$